

# Incomplete Markets II

## Stationary Equilibrium, Aiyagari (1994), and Capital Taxation

Macroeconomics — Lecture Notes

### 1. Stationary Rational Expectations Equilibrium (Huggett, 1993)

#### Stationary Rational Expectations Equilibrium

Also called **recursive stationary equilibrium**. It consists of:

- a) A **value function**  $V : S \rightarrow \mathbb{R}$
- b) **Policy functions**  $c : S \rightarrow \mathbb{R}$  and  $a' : S \rightarrow \mathbb{R}$
- c) An **interest rate**  $r^*$
- d) A **stationary distribution**  $\mu^*$

such that the following conditions hold simultaneously.

#### 1.1. Conditions of the Equilibrium

(i)  **$V$  is the associated value function** —  $V$  solves the household's Bellman equation given  $r^*$ :

$$V(a_i, y_i) = \max_{c_i, a'_i} \left\{ u(c_i) + \beta \sum_{y'_i} \pi(y'_i, y_i) V(a'_i, y'_i) \right\}$$

subject to  $c_i + a'_i = (1 + r^*)a_i + y_i$  and  $a'_i \geq -b$ .

(ii) **Asset market clears** at interest rate  $r^*$ :

$$\int_{A \times Y} a'(a_i, y_i) d\mu^* = 0$$

Assets are in **zero net supply** — the economy is an endowment economy where households lend to and borrow from each other. Total savings equal total debt.

(iii) **Goods market clears**:

$$\int_{A \times Y} c(a_i, y_i) d\mu^* = \sum_{y_i} y_i \Pi^*(y)$$

Aggregate consumption equals aggregate endowment.

(iv)  **$\mu^*$  is the fixed point** of the law of motion  $(*)$ , i.e., for all  $\mathcal{A} \times \mathcal{Y}$ :

$$\mu^*(\mathcal{A} \times \mathcal{Y}) = \int_{A \times Y} Q(\mathcal{A} \times \mathcal{Y} \mid a_i, y_i) d\mu^*$$

The distribution of households over states reproduces itself each period — it is stationary.

## 2. Existence and Uniqueness of Equilibrium

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### 2.1. Aggregate Demand for Assets

Define the **aggregate asset demand** as a function of  $r$ :

$$A(r) = \int_{A \times Y} a'(a_i, y_i; r) d\mu_r^*$$

where both  $a'(\cdot)$  and  $\mu^*$  depend explicitly on  $r$ . The equilibrium condition is  $A(r^*) = 0$ .

### 2.2. Key Properties of $A(r)$

- $A\left(\frac{1}{\beta} - 1\right) \rightarrow +\infty$ : when  $r \rightarrow \frac{1}{\beta} - 1$ , saving is so attractive that households want to accumulate infinite wealth.
- $A(-1) = -b$ : when  $r = -1$ , the gross return  $(1 + r) = 0$  destroys savings, so every household borrows to the maximum limit  $-b$ .

#### Key Result

Since  $A(-1) = -b < 0$  and  $A\left(\frac{1}{\beta} - 1\right) \rightarrow +\infty > 0$ , by the **Intermediate Value Theorem** there exists at least one  $r^* \in \left(-1, \frac{1}{\beta} - 1\right)$  such that  $A(r^*) = 0$ . This guarantees **existence** of the equilibrium.

### 2.3. Continuity and Uniqueness

- **Berge's Maximum Theorem**  $\Rightarrow a'(\cdot)$  is continuous in  $r$ .
- For  $\mu^*$  to be continuous in  $r$ , we need a finite upper bound on wealth, which requires  $\beta(1 + r) < 1$ .
- **Uniqueness** of  $\mu^*$  is guaranteed if  $Q(\cdot)$  satisfies the **monotone mixing condition**.
- **Uniqueness** of  $r^*$  requires  $A(r)$  to be strictly increasing in  $r$  — hard to prove in general.

## 3. The Risk-Free Rate Puzzle

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#### The Puzzle

In the data we simultaneously observe: (i) a **high equity premium** and (ii) a **low risk-free rate**. Standard representative agent models cannot reconcile both facts.

The tension is as follows. Explaining a high equity premium requires **high risk aversion**  $\sigma$ . But high risk aversion implies a strong consumption smoothing motive and a strong desire to borrow against future income — which pushes the risk-free rate **up**. This contradicts the low rate observed in the data.

### Key Result

**Huggett (1993)** shows that the presence of idiosyncratic risk helps resolve this puzzle. Uninsurable income shocks create a **precautionary saving motive** — households save more as a buffer against future bad shocks. This extra saving pushes the equilibrium risk-free rate **down**, reconciling high risk aversion with a low risk-free rate.

## 4. A Production Economy: Aiyagari (1994)

### 4.1. Key Differences from Huggett

Aiyagari (1994) extends Huggett's framework by replacing the endowment economy with a **production economy**. The key changes are:

|                       | Huggett (1993)          | Aiyagari (1994)                     |
|-----------------------|-------------------------|-------------------------------------|
| Economy type          | Endowment               | Production                          |
| Asset $a$             | Bond between households | Financial claim to physical capital |
| Asset market clearing | $A(r^*) = 0$            | $A(r^*) = K(r^*)$                   |
| Goods market clearing | $C = Y$                 | $C + \delta K = F(K, L)$            |

### 4.2. Setup

- Households supply **1 unit of labour inelastically** each period.
- $y_i$ : **efficiency units of labour** — the idiosyncratic productivity shock.
- $a_i$ : **financial claims to physical capital** — households own the capital used by the firm.
- There is a **competitive representative firm** with a CRS production function  $F(K, L)$ .
- $w$ : wage rate;  $r$ : interest rate — both determined in equilibrium.

### 4.3. Equilibrium Conditions

(i) **Asset market clears:**

$$\int_{A \times Y} a'(a_i, y_i; r^*) d\mu_{r^*}^* = K(r^*)$$

Household savings finance the physical capital stock of the economy.

(ii) **Goods market clears:**

$$\int_{A \times Y} c(a_i, y_i; r^*) d\mu_{r^*}^* + \delta K(r^*) = F(K, L)$$

Aggregate consumption plus capital depreciation equals total output.

(iii)  $\mu^*$  is the fixed point of the law of motion:

$$\mu_{r^*}^*(\mathcal{A} \times \mathcal{Y}) = \int_{\mathcal{A} \times \mathcal{Y}} Q(\mathcal{A} \times \mathcal{Y} \mid a_i, y_i) d\mu_{r^*}^*$$

#### 4.4. Firm's Demand for Capital

The competitive firm equates the marginal product of capital to its cost:

$$r + \delta = F_K(K, L)$$

This pins down  $K(r)$  — the firm's capital demand as a function of  $r$ . Key properties:

- $K(r)$  is **continuous and strictly decreasing** in  $r$ .
- As  $r \rightarrow -\delta$ :  $K \rightarrow \infty$  (capital is essentially free).
- As  $r \rightarrow \infty$ :  $K \rightarrow 0$  (capital is too expensive).

The equilibrium  $r^*$  is where household asset supply meets firm capital demand:  $A(r^*) = K(r^*)$ .

## 5. Capital Taxation: Chamley–Judd vs. Aiyagari

### The Question

Should the government tax capital in the long run?

#### 5.1. Chamley–Judd (1986): $\tau_k = 0$

In a **complete markets, representative agent** economy, the optimal long-run capital tax is **zero**. The key distortion is visible in the Euler equation. Without taxes:

$$u'(c_t) = \beta(1 + r) u'(c_{t+1})$$

With a capital tax  $\tau$ , the net return becomes  $r(1 - \tau)$ :

$$u'(c_t) = \beta(1 + r(1 - \tau)) u'(c_{t+1})$$

This introduces a **wedge** between the social return  $r$  and the private return  $r(1 - \tau)$ :

$$\underbrace{r}_{\text{social return}} \neq \underbrace{r(1 - \tau)}_{\text{private return}}$$

Households see a lower return and save **less** than is socially optimal. In the long run, any  $\tau > 0$  accumulates this distortion indefinitely — the government prefers to raise revenue through labour taxes instead.

#### 5.2. Aiyagari (1995): $\tau_k > 0$

With **incomplete markets and heterogeneous agents**, the optimal long-run capital tax is **positive**. The key insight: households facing uninsurable idiosyncratic risk accumulate **too much** capital due to precautionary saving — more than the social optimum. A capital tax corrects this over-accumulation.

### Key Result

**The punchline:** incomplete markets change everything. The Chamley–Judd zero-tax result breaks down once households cannot perfectly insure against income risk.

|                     | Markets    | Agents         | Optimal $\tau_k$ |
|---------------------|------------|----------------|------------------|
| Chamley–Judd (1986) | Complete   | Representative | $\tau_k = 0$     |
| Aiyagari (1995)     | Incomplete | Heterogeneous  | $\tau_k > 0$     |

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End of Incomplete Markets II Notes